Analysis and Design of Load-Independent Class-E Zero-Voltage Switching Power Oscillator

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***Abstract*—This paper proposes the load-independent class-E Zero-Voltage Switching (ZVS) power oscillator. Conventional class-E power oscillator has a load-dependent phase shift, which hinders sustained oscillation under varying load conditions. Hence, we introduce the class-E power amplifier that ensures the load-independent phase shift in the output current. By incorporating it into an LCLC filter with a resonant-capacitor feedback network, the load-independent self-oscillation is realized. The proposed power oscillator satisfies a phase-shift requirement for sustained oscillation regardless of the load resistance. Furthermore, the proposed circuit exhibits load independence in ZVS, output current, and gate-drive voltage. This paper provides a thorough analysis of the proposed power oscillator, covering its operating principle, power loss prediction, design, and limitations. This paper also gives two design examples with 0.8 MHz and 6.78 MHz oscillation frequencies. In the experiment, the prototype power oscillators achieved 93.8 % and 87.4 % power-conversion efficiencies, respectively. The validity and effectiveness of the proposed power oscillator are demonstrated from the experimental verifications.**

***Index Terms*—Class-E power oscillator, constant current, load-independent, zero-voltage switching.**

I. Introduction

THE wide-bandgap power semiconductor devices have enabled the implementation of high-frequency circuits operating beyond MHz [1], [2]. High-frequency operation reduces the physical size of the passive components, increasing the power density of the overall circuit. One major challenge in high-frequency operation is the growing switching losses. In order to reduce the switching losses, the circuit needs to achieve Zero-Voltage Switching (ZVS).

The class-E power oscillator [3] is one of the AC generators achieving ZVS. The class-E power oscillator consists of the class-E power amplifier and a passive feedback network. The gate-drive voltage of the power oscillator is generated from the output AC voltage. Meanwhile, the output voltage has a certain phase shift from the gate-drive voltage. The phase shift is compensated by the feedback network to realize the sustained oscillation. In addition, the class-E power amplifier requires appropriate component-value selection to achieve ZVS [4]. Under a fine-tuning condition, the class-E power oscillator achieves high-efficiency operation. However, the class-E power oscillator is highly sensitive to load variations due to the load-dependent phase shift. The load variations cause efficiency deterioration due to the non-ZVS and fluctuations in output. Even worse, the oscillation may stop. Various types of power oscillators, such as class-E [3], [5], [6], [7], [8], [9], [10], [11], class-E/F_3 [12], [13], class-Φ_3 [14], injection-locked class-E [15], [16], and class-E_M [17] power oscillators, have been proposed so far. Nevertheless, all these power oscillators achieve optimal performance only within a narrow range of load variations. To address the sensitivity issue against load variations, load-independent power oscillators have been proposed [18], [19], [20], [21]. These power oscillators inherently achieve Zero-Current Switching (ZCS) and constant output-voltage amplitude against load variations. The robustness against load variations is obtained by using a specific set of component values [22], [23]. However, the ZCS circuits cannot absorb the drain-source parasitic capacitance of the switching device. The parasitic capacitance causes excessive switch-voltage stress and power loss due to non-ZVS [24], [25]. Therefore, the ZVS power oscillators are preferable for high-frequency operation in the multi-MHz range.

This paper proposes the load-independent class-E ZVS power oscillator. The contribution of this paper can be summarized as follows.

- The proposed power oscillator is the first power oscillator that achieves load-independent ZVS. The proposed power oscillator also ensures constant amplitudes in output current and gate-drive voltage against load variations.
- The proposed circuit topology inherently absorbs the input capacitance of the switch. This feature allows stable sinusoidal driving even at high-frequency operation.
- Closed-form design equations are presented. The design equations allow for designing the proposed power oscillator quickly and intuitively with any specifications.
- Analytical design conditions and power-loss predictions for the proposed power oscillator are provided, which are validated by the experiment.

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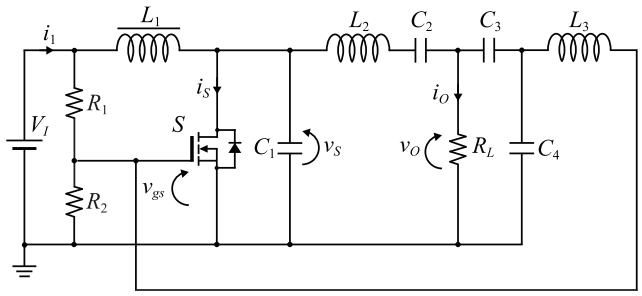


Fig. 1. Circuit schematic of the conventional class-E power oscillator.

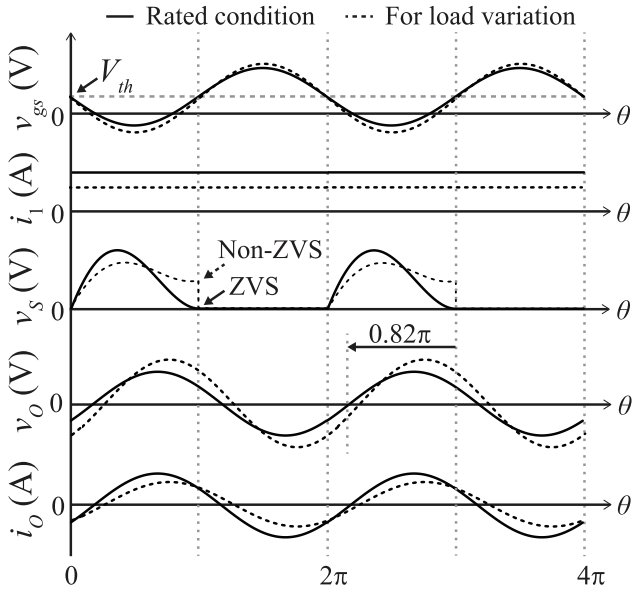


Fig. 2. Example waveforms of the conventional class-E power oscillator.

The proposed power oscillator is derived from the class-E power amplifier. The class-E power amplifier is designed to have a load-independent phase shift in the output current. To make a closed loop, an additional phase shift for the output current is provided by introducing the LCLC filter with the resonant-capacitor feedback network. Consequently, the proposed topology realizes the load-independent operation in the power oscillator. The experimental verifications were carried out at 0.8 MHz and 6.78 MHz oscillation frequencies. The measured waveforms were quantitatively agreed with the analytical one, confirming the validity of the experiment.

II. Conventional Class-E Power Oscillator

Fig. 1 illustrates the circuit schematic of the conventional class-E power oscillator [3], comprising a class-E power amplifier and a feedback network. The class-E power oscillator works without the gate-drive circuits. Namely, a crystal oscillator, gate driver, and additional power supplies are not required.

Fig. 2 illustrates the example waveforms of the class-E power oscillator, where $\theta = \omega_S t = 2\pi f_S t$ represents the angular time, ω_S indicates the angular oscillation frequency, and f_S denotes the oscillation frequency. The switch is driven by the excitation of the biased sinusoidal voltage v_{gs}, which is generated using the output AC voltage v_O. To induce the sustained oscillation, the total phase shift in the closed loop must be an integer multiple of 2π. Meanwhile, the nominal phase shift of the output voltage is 0.82π in the class-E power amplifier [4], as shown in Fig. 2. In order to compensate for the phase shift between the output and the gate-drive voltages, the feedback network C_3-C_4-L_3 needs to provide an appropriate phase shift. In addition, the class-E power oscillator features to achieve ZVS at the rated condition. Achieving ZVS requires the careful tuning of the components.

One of the major issues in the class-E power oscillator is its sensitivity to load variations. When the load variations occur, the class-E power oscillator is unable to maintain ZVS, and both the output voltage and current fluctuate, as shown in Fig. 2. The high sensitivity to load variations is attributable to the features in the class-E power amplifier. The class-E power amplifier changes the phase shift of the output voltage against load variations [26], [27]. Consequently, the class-E power oscillator forcibly changes the oscillation frequency to satisfy the phase-shift requirement against load variations. Therefore, even slight load variations can seriously affect the overall performance of the power oscillator.

III. Class-E Amplifier Wtih Load-Independent Phase Shift

As discussed in Section II, the sensitivity of the class-E power oscillator to load variations stems from the load-dependent phase shift in the class-E power amplifier. To address this issue, we aim to design the class-E power amplifier that maintains a constant phase shift in the output current against load variations. The analysis is performed for the class-E power amplifier with finite-DC feed inductance [28] under the following assumptions.

(i) The series-resonant filter functions as an ideal band-pass filter. The band-pass filter only passes a fundamental component. Thus, the output current can be expressed as a pure sinusoidal waveform as

$$i_{eq}(\theta) = \frac{V_{eq}}{R_{eq}} \sin(\theta + \varphi), \tag{1}$$

where V_{eq} is the amplitude of the voltage across R_{eq}, and φ is the phase shift of the output current.

(ii) The inductor L_2 is divided into an extra-inductive component L_X and a resonant component L_R.

(iii) The switching device functions as an ideal switch, turning OFF during $0 \le \theta < \pi$ and turning ON during $\pi \le \theta < 2\pi$. Additionally, it possesses zero ON resistance, zero gate resistance, and infinite OFF resistance.

(iv) All the passive components operate linearly without parasitic components.

From the assumptions above, the equivalent circuit model of the class-E power amplifier is given as shown in Fig. 3.

Since this section focuses on the design conditions for the constant phase shift, the detailed procedure for obtaining the waveform equations is presented in the Appendix. The

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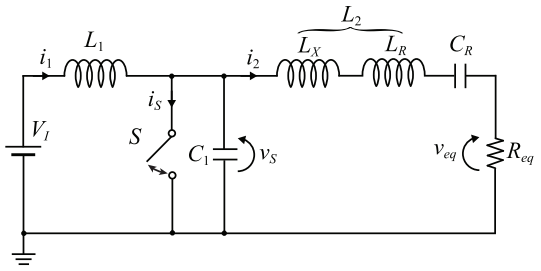


Fig. 3. Equivalent circuit model of the class-E power amplifier.

waveform equations for input current and switch voltage are given as shown in (2) and (3), at the bottom of the page, where the normalized resonant frequency ω_1 is given as

$$\omega_1 = \frac{1}{\omega_S \sqrt{L_1 C_1}}. \tag{4}$$

Also, α and β are defined as

$$\alpha = \frac{1}{\omega_S C_1 \omega_1^2}\left[\frac{V_{eq}\cos\varphi}{R_{eq}(\omega_1^2-1)} - \frac{V_I}{\omega_S L_1}\right], \tag{5}$$

$$\beta = \frac{\frac{V_I}{\omega_S L_1}\left[\pi - \frac{\sin(\pi\omega_1)}{\omega_1}\right] - \frac{\omega_1 V_{eq}[\omega_1 \sin\varphi + \sin(\pi\omega_1)\cos\varphi]}{R_{eq}(\omega_1^2-1)}}{\omega_S C_1 \omega_1 [1-\cos(\pi\omega_1)]}, \tag{6}$$

respectively.

For designing the class-E power amplifier, we impose the turn-ON ZVS condition as

$$v_S(\pi) = 0. \tag{7}$$

From (2) and (7), we obtain

$$\alpha\cos(\pi\omega_1) + \beta\sin(\pi\omega_1) + V_I + \frac{\omega_S L_1 \omega_1^2 V_{eq}\cos\varphi}{R_{eq}(\omega_1^2-1)} = 0. \tag{8}$$

In order to ensure the load-independent phase shift in the output current, the following condition needs to be satisfied.

$$\frac{\partial\varphi}{\partial R_{eq}} = 0. \tag{9}$$

Since (8) is the implicit function of φ and R_{eq}, $\partial\varphi/\partial R_{eq}$ can be obtained by differentiating both sides of (8) with respect to R_{eq} as

$$\frac{\partial\varphi}{\partial R_{eq}} = \frac{\omega_1 \tan\varphi}{R_{eq}\left[\omega_1 - \tan\left(\frac{\pi\omega_1}{2}\right)\tan\varphi\right]}. \tag{10}$$

From (9) and (10), we obtain

$$\varphi = 0. \tag{11}$$

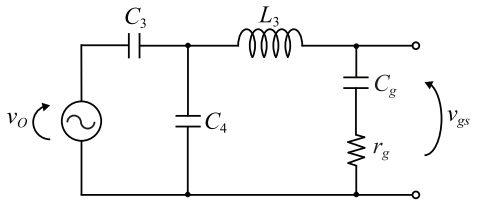


Fig. 4. Circuit model of the feedback network in the conventional class-E power oscillator.

It can be seen from (11) that the phase shift between gate-drive voltage and output current always becomes π for designing the class-E power amplifier with the load-independent phase shift.

From (5), (6), (8), and (11), the design condition can be obtained as

$$\tan\left(\frac{\pi\omega_1}{2}\right) + \frac{\pi\omega_1}{2} = 0, \tag{12}$$

which can be numerically solved as

$$\omega_1 \approx 1.292. \tag{13}$$

It is seen from (13) that the input inductor L_1 needs to resonate with the shunt capacitor C_1 at a frequency slightly above the switching frequency.

IV. Load-Independent Class-E Power Oscillator

From the analysis given in Section III, it is found that the class-E power amplifier can have the load-independent phase shift in the output current under a certain design condition. Therefore, this section attempts to incorporate the class-E power amplifier into the power oscillator.

A. Investigation in Conventional Circuit Topology

We investigate the feasibility of adopting the load-independent design using the conventional class-E power oscillator topology introduced in Section II.

Fig. 4 shows the circuit model of the feedback network in the conventional class-E power oscillator, where C_g and r_g are the gate parasitic capacitance and gate resistance of the switching device, respectively. The phase difference between the output voltage v_O and gate-drive voltage v_{gs} is calculated as

$$\varphi_O - \varphi_{gs} = \arctan\left\{\frac{\omega C_g\left[\omega L_3 - \frac{1}{\omega(C_3+C_4)}\right]}{\frac{1}{\omega(C_3+C_4)r_g} + \frac{\omega L_3 - \frac{1}{\omega C_g}}{r_g} - \omega C_g r_g}\right\}, \tag{14}$$

$$v_S(\theta) = \begin{cases} \alpha\cos(\omega_1\theta) + \beta\sin(\omega_1\theta) + V_I - \dfrac{\omega_S L_1 \omega_1^2 V_{eq}\cos(\theta+\varphi)}{R_{eq}(\omega_1^2-1)}, & (0 \le \theta < \pi) \\ 0, & (\pi \le \theta < 2\pi) \end{cases} \tag{2}$$

$$i_1(\theta) = \begin{cases} -\dfrac{\omega_1[\alpha\sin(\omega_1\theta) - \beta\cos(\omega_1\theta)]}{\omega_S L_1} + \dfrac{\omega_1^2 V_{eq}\sin(\theta+\varphi)}{R_{eq}(\omega_1^2-1)}, & (0 \le \theta < \pi) \\ \dfrac{V_I(\theta-\pi) - \omega_1[\alpha\sin(\pi\omega_1) - \beta\cos(\pi\omega_1)]}{\omega_S L_1}, & (\pi \le \theta < 2\pi) \end{cases} \tag{3}$$

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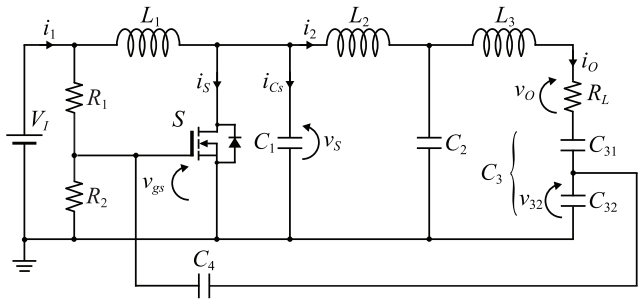


Fig. 5. Circuit schematic of the proposed load-independent class-E power oscillator.

where φ_O and φ_{gs} are the phase shifts of the output and gate-drive voltages, $\omega = 2\pi f$ is the angular frequency, and f is the frequency. For applying the load-independent design, the feedback network must provide π phase shift. However, there is no appropriate solution that satisfies $\varphi_O - \varphi_{gs} = \pm\pi$ in (14). This means that the feedback network of the conventional class-E power oscillator cannot compensate for the phase shift. Therefore, the topological change is necessary to incorporate the load-independent design into the power oscillator.

B. Proposed Circuit

Fig. 5 shows the circuit schematic of the proposed load-independent class-E power oscillator. The proposed power oscillator is composed of input DC-voltage source V_I, switching device S, finite inductor L_1, shunt capacitor C_1, LCLC filter L_2-C_2-L_3-C_3, load resistance R_L, DC-blocking capacitor C_4, and voltage-divide resistances R_1 and R_2. The capacitor C_3 is divided into C_{31} and C_{32} to adjust the amplitude of the gate-drive voltage.

This paper introduces an LCLC filter with the resonant-capacitor feedback network to provide an additional phase shift to the class-E power amplifier. Fig. 6 shows typical waveforms of the proposed power oscillator. The LCLC filter provides a $\pi/2$ phase shift between the currents i_2 and i_O under a specific resonant condition. Also, the phase shift of the voltage across the capacitor C_{32} leads $\pi/2$ against the output current i_O. Consequently, the voltage v_{32} becomes in phase with the gate-drive voltage, and the power oscillator satisfies the phase-shift requirement. Most notably, these phase shifts are fixed and independent of the load resistance. Besides, the input impedance of the LCLC filter becomes purely resistive in this particular resonant condition. The resistive impedance does not interfere with the load-independent operation of the power oscillator.

We can also see the switching process of the proposed power oscillator from Fig. 6. After turn-OFF, current flows into the shunt capacitor, causing the switch voltage to gradually rise. When the switch voltage reaches its peak, the shunt capacitor begins to discharge, and the switch voltage decreases. At the turn-ON moment, the switch voltage becomes zero. Namely, the proposed power oscillator ensures the ZVS without body-diode conduction.

The proposed power oscillator exhibits the load-independent characteristics in ZVS, output current, and gate-drive voltage.

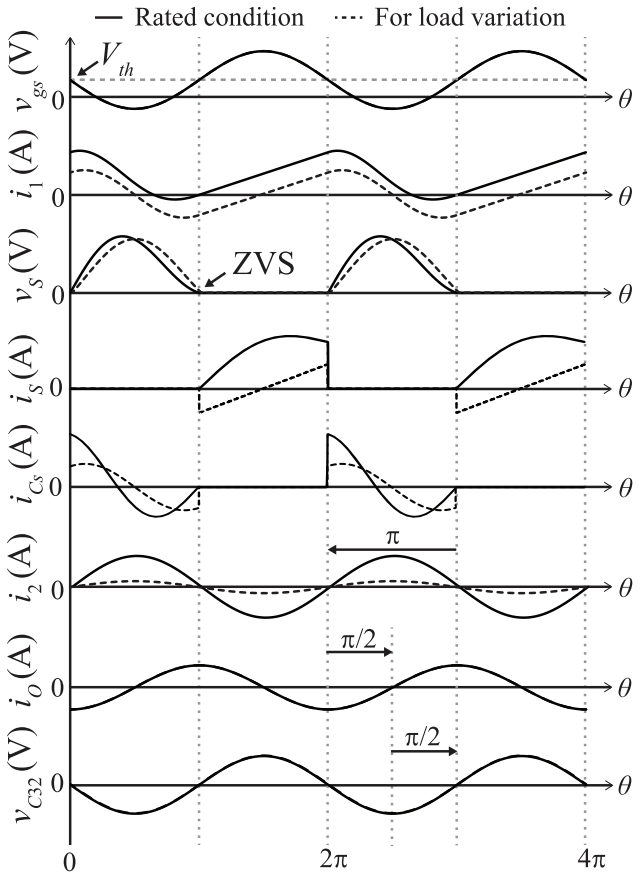


Fig. 6. Typical waveforms of the load-independent class-E power oscillator.

The constant output current is useful in applications such as transmitters of the WPT systems [29]. Additionally, the constant current characteristic makes the amplitude of the voltage v_{32} also constant. As a result, the amplitude of the gate-drive voltage becomes independent of the load resistance, which further improves the stability of the oscillation.

Another critical advantage of the proposed circuit topology is to absorb the gate parasitic capacitance of the switching device by the capacitor C_{32}. In the conventional circuit topology, the nonlinearity of the gate capacitance distorts the gate-drive voltage [9]. On the other hand, the proposed power oscillator can generate a pure sinusoidal gate-drive voltage. The device tolerance of the gate capacitance can also be absorbed. As a result, the proposed power oscillator works properly even without the gate driver.

V. Circuit Analysis of Proposed Power Oscillator

This section provides the circuit analysis for the proposed load-independent class-E power oscillator. We analytically clarify the design conditions of the LCLC filter. Furthermore, the load-independent characteristics of the proposed power oscillator are also confirmed through the analysis.

Fig. 7 shows the analytical circuit model of the proposed power oscillator. In addition to the assumption given in Section III, we assume that the capacitor C_4 works as a DC blocking and is large enough to be regarded as a DC-voltage

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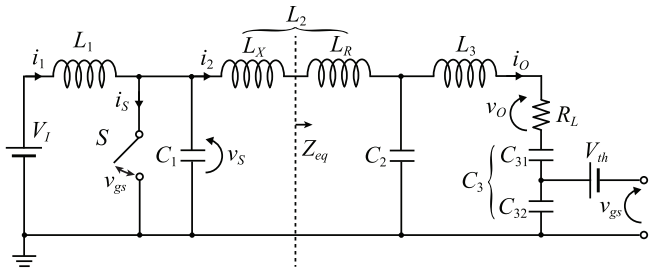


Fig. 7. Analytical circuit model of the proposed power oscillator.

source V_{th}. Besides, the resistances R_1 and R_2 are large enough so that these components are regarded not conducting current.

A. LCLC Filter

The input impedance of the LCLC filter is obtained as

$$Z_{eq} = \frac{R_L}{\omega^2 C_2^2 (R_L^2 + H_2^2)} + j\left[H_1 - \frac{H_2}{\omega^2 C_2^2 (R_L^2 + H_2^2)}\right], \quad (15)$$

where H_1 and H_2 are defined as

$$H_1 = \omega L_R - \frac{1}{\omega C_2}, \quad (16)$$

and

$$H_2 = \omega L_3 - \frac{1}{\omega C_2} - \frac{1}{\omega C_3}, \quad (17)$$

respectively.

For ensuring the load-independent operation, the input impedance of the LCLC filter is designed to be purely resistive. Namely, the resonant conditions should be $H_1 = H_2 = 0$, which determines the angular oscillation frequency ω_S as

$$\omega_S = 2\pi f_S = \frac{1}{\sqrt{L_R C_2}} = \sqrt{\frac{1}{L_3}\left(\frac{1}{C_2} + \frac{1}{C_3}\right)}. \quad (18)$$

When (18) is satisfied, the input impedance in (15) becomes

$$Z_{eq} = R_{eq} = \frac{1}{\omega_S^2 C_2^2 R_L}. \quad (19)$$

We can see from (16)–(18) that the resonant conditions hold regardless of the load resistance R_L. Therefore, (19) indicates that the variations of load resistance can be seen as resistive variations for the class-E power amplifier.

The voltage gain G_f and phase shift φ_f between v_{eq} and v_O can be obtained as

$$G_f = \frac{V_O}{V_{eq}} = \sqrt{\frac{\omega^2 C_2^2 R_L (R_L^2 + H_2^2)}{\sqrt{R_L^2 + [\omega^2 C_2^2 (R_L^2 + H_2^2) H_1 - H_2]^2}}}, \quad (20)$$

and

$$\varphi_f = \begin{cases} \arctan\left(\dfrac{1}{\omega^2 C_1^2 R_L H_1} - \dfrac{H_2}{R_L}\right) & \text{for} \quad H_1 < 0, \\ -\pi + \arctan\left(\dfrac{1}{\omega^2 C_1^2 R_L H_1} - \dfrac{H_2}{R_L}\right) & \text{for} \quad H_1 \geq 0, \end{cases} \quad (21)$$

respectively, where V_{eq} and V_O are the amplitudes of the input and output voltages of the LCLC filter. From (20) and (21), the voltage gain and phase shift under the resonant condition in (18) are derived as

$$G_f = \omega_S C_2 R_L = \frac{R_L}{\omega_S L_R}, \quad (22)$$

and

$$\varphi_f = -\frac{\pi}{2}, \quad (23)$$

respectively.

From (1), (11), (22), and (23), the voltage across capacitor C_{32} can be expressed as

$$v_{32}(\theta) = -\frac{C_2 V_{eq}}{C_{32}} \sin\theta, \quad (24)$$

Accordingly, the gate-drive voltage v_{gs} can be obtained as

$$v_{gs}(\theta) = -\frac{C_2 V_{eq}}{C_{32}} \sin\theta + V_{th}. \quad (25)$$

B. Load-Independent Characteristics

The proposed power oscillator exhibits the load independence in ZVS, output current, and gate-drive voltage, which can be confirmed through the analysis. From assumption (i), the band-pass filter only passes the fundamental frequency component. Accordingly, the amplitude of the voltage across the equivalent resistance R_{eq} and the extra inductance L_X can be obtained from the first harmonic term of the Fourier Series Expansion as

$$V_{eq} = \frac{1}{\pi}\int_0^{2\pi} v_S(\theta) \sin(\theta + \varphi) d\theta, \quad (26)$$

and

$$\frac{\omega_S L_X V_{eq}}{R_{eq}} = \frac{1}{\pi}\int_0^{2\pi} v_S(\theta) \cos(\theta + \varphi) d\theta, \quad (27)$$

respectively.

From (2), (5), (6), (11), and (26), the amplitude of the voltage across the equivalent resistance R_{eq} becomes

$$V_{eq} = \frac{\omega_1^3 V_I}{(1 - \omega_1^2)\tan\left(\frac{\pi\omega_1}{2}\right)} \approx 1.590 V_I. \quad (28)$$

We can see from (28) that the amplitude of the voltage v_{eq} is independent of the load resistance. Therefore, the gate-drive voltage in (25) is also independent of the load resistance. Additionally, the amplitude of the output current I_O can be derived from (22) and (28) as

$$I_O = \frac{\omega_1^3 V_I}{\omega_S L_1 (1 - \omega_1^2)\tan\left(\frac{\pi\omega_1}{2}\right)} \approx \frac{1.590 V_I}{\omega_S L_1}. \quad (29)$$

which is also unaffected by the load resistance.

The fact that the proposed power oscillator achieves the ZVS regardless of the load resistance can be verified using (2), (5), and (6) as

$$\frac{\partial v_S(\pi)}{\partial R_{eq}} = \frac{\partial}{\partial R_{eq}}\left\{\left[2 + \frac{\pi\omega_1 \sin(\pi\omega_1)}{1 - \cos(\pi\omega_1)}\right] V_I\right\} = 0. \quad (30)$$

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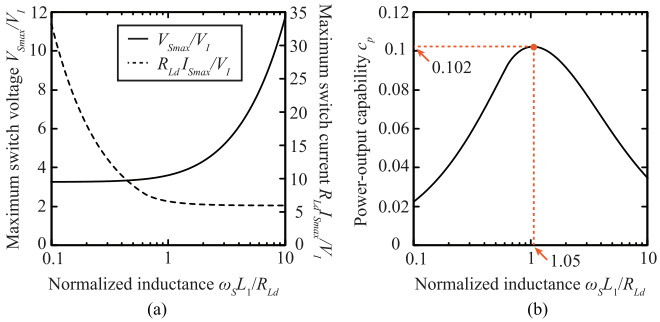


Fig. 8. Circuit characteristics as functions of normalized inductance $\omega_S L_1/R_{Ld}$. (a) Normalized maximum switch voltage and current. (b) Power-output capability c_p.

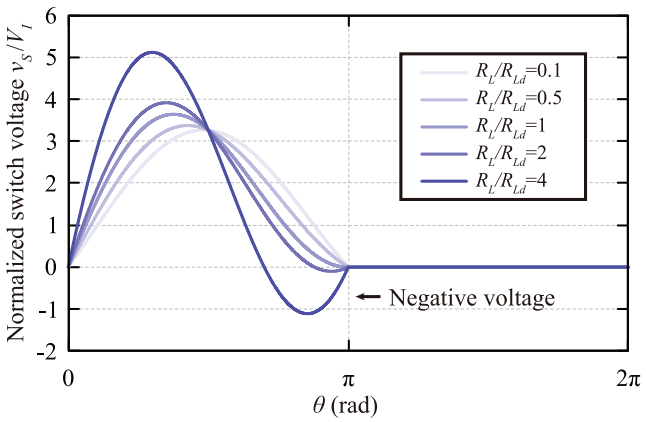


Fig. 9. The waveforms of the normalized switch voltage of the load-independent class-E power oscillator.

VI. Circuit Design of Proposed Power Oscillator

This section provides the design considerations and closed-form design equations of the proposed power oscillator. The design equations determine the component values uniquely from the specifications, allowing intuitive and quick circuit design.

A. Optimization of Power-Output Capability

The proposed power oscillator can be designed with arbitrary input inductance L_1. Therefore, a reasonable value of the inductance should be explored. Fig. 8 shows the circuit characteristics as functions of normalized inductance $\omega_S L_1/R_{Ld}$, where R_{Ld} is the value of load resistance at the design point. The power-output capability is calculated as

$$c_p = \frac{P_O}{V_{Smax} I_{Smax}} = \frac{V_{eq}^2}{2R_{eq} V_{Smax} I_{Smax}}, \tag{31}$$

where P_O is the output power, and V_{Smax} and I_{Smax} are peak values of the switch voltage and current, respectively. It can be seen from Fig. 8(a) that the maximum switch voltage monotonically decreases as the increase of L_1. Meanwhile, the maximum switch current monotonically decreases as the increase of L_1. Thus, we can find from (28) and (31) that certain L_1 maximizes the power-output capability. We can see from Fig. 8(b) that the power-output capability is maximized to $c_p = 1.02$ at $\omega_S L_1/R_{Ld} = 1.05$.

B. Limitation of Load-Independent Operating Range

Fig. 9 shows the waveforms of the normalized switch voltage of the load-independent class-E power oscillator. The load-independent turn-ON ZVS is theoretically guaranteed for any load resistance. Meanwhile, the shape of the switch voltage changes depending on the load resistance, making the voltage negative for $R_L/R_{Ld} > 1$, as shown in Fig. 9. As the analysis given in this paper does not consider the body diode of the switching device, the negative voltage at the turn-OFF period may be clamped by the reverse conduction in the real devices. Consequently, the constant output current characteristic will be lost for $R_L/R_{Ld} > 1$. Therefore, the load range of the proposed power oscillator should be designed for $R_L/R_{Ld} \leq 1$ to obtain the load-independent output current.

C. Requirement for Self-Starting Oscillation

In the proposed power oscillator, once the input voltage is applied, the capacitor C_4 stores electric charge. When the voltage across C_4 exceeds the threshold voltage of the switch V_{th}, the switch turns on. This event triggers resonance, initiating the self-oscillation [10].

The voltage across C_4, specified as V_4, can be given as

$$V_4 = \frac{R_2 V_I}{R_1 + R_2}. \tag{32}$$

The self-starting oscillation can be ensured by satisfying $V_4 \geq V_{th}$. However, there can be variations in the threshold voltage of the switch. When $V_4 < V_{th}$, the self-oscillation may not be ensured. Even when oscillation occurs, the ON-duty ratio becomes smaller than 50 %, which can lead to non-ZVS. Therefore, it is recommended to design V_4 slightly higher than V_{th}.

D. Design Equation

The component values of the proposed power oscillator are given by the closed-form design equations. We assume that the specifications for the oscillation frequency f_S, input voltage V_I, load resistance at the design point R_{Ld}, amplitude of the gate-drive voltage V_{gs}, and normalized inductances $Q_2 = \omega L_2/R_{Ld}$ and $Q_3 = \omega L_3/R_{Ld}$ are given beforehand.

From the discussion in Section VI-A, L_1 is determined to maximize power-output capability as

$$L_1 = \frac{0.167\ R_{Ld}}{f_S}. \tag{33}$$

From (13) and (33), the shunt capacitance C_1 is obtained as

$$C_1 = \frac{0.0909}{f_S R_{Ld}}. \tag{34}$$

From the specifications, we obtain

$$L_2 = \frac{Q_2 R_{Ld}}{2\pi f_S}, \tag{35}$$

and

$$L_3 = \frac{Q_3 R_{Ld}}{2\pi f_S}. \tag{36}$$

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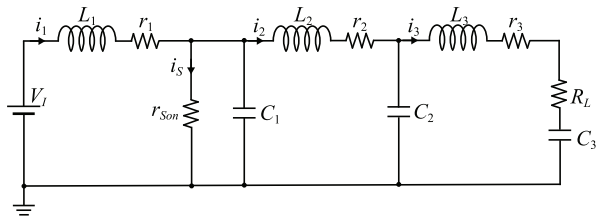


Fig. 10. Circuit model of the proposed power oscillator for power loss calculation.

From (2), (5), (6), (11), and (27), the ratio of L_X and L_1 is determined uniquely as

$$\frac{L_X}{L_1} = -\frac{2\omega_1^3}{\pi(\omega_1^2-1)^2\tan\left(\frac{\pi\omega_1}{2}\right)} - \frac{\omega_1}{2(\omega_1^2-1)} \approx 0.2662. \quad (37)$$

Therefore, the extra inductance L_X is determined from (33) and (37) as

$$L_X = \frac{0.0445R_{Ld}}{f_S}. \quad (38)$$

From (16) and (18), the capacitance C_2 becomes

$$C_2 = \frac{0.0253}{f_S^2(L_2 - L_X)}. \quad (39)$$

From (17) and (18), the capacitance C_3 is

$$C_3 = \frac{0.0253}{f_S^2(L_3 - L_2 + L_X)}. \quad (40)$$

From (24) and (28), we obtain

$$C_{32} = \frac{0.0403V_I}{f_S^2 L_2 V_{gs}}. \quad (41)$$

As the capacitances C_{31} and C_{32} are connected in series, the capacitance C_{31} is obtained from (40) and (41) as

$$C_{31} = \frac{C_{32}C_3}{C_{32} - C_3}. \quad (42)$$

VII. Power-Loss Prediction

This section provides the power loss analysis for the proposed power oscillator. Fig. 10 shows the circuit model of the proposed power oscillator for power loss calculation, which includes the Equivalent Series Resistances (ESRs) of the inductors and ON-resistance of the switching device. In order to predict the power loss in each component, we utilize the waveform equations derived in Section V. Here, we assume that the shape of the waveforms does not change even with the ESRs and the ON-resistance.

The power loss caused by the ESRs of the inductors L_1, L_2, and L_3 can be expressed as

$$P_{L1} = \frac{r_1}{2\pi}\int_0^{2\pi} i_1(\theta)^2 d\theta, \quad (43)$$

$$P_{L2} = \frac{r_2}{2\pi}\int_0^{2\pi} i_2(\theta)^2 d\theta = \frac{r_2 V_{eq}^2}{2R_{eq}^2}, \quad (44)$$

and

$$P_{L3} = \frac{r_3}{2\pi}\int_0^{2\pi} i_3(\theta)^2 d\theta = \frac{r_3 I_O^2}{2}, \quad (45)$$

respectively.

Since the proposed power oscillator theoretically achieves the exact ZVS regardless of the load resistance, power losses due to the ON resistance and the gate drive are only taken into account in the switching device. The power loss caused by the ON-resistance can be expressed as

$$P_{Son} = \frac{r_{Son}}{2\pi}\int_0^{2\pi} i_S(\theta)^2 d\theta = \frac{r_{Son}}{2\pi}\int_\pi^{2\pi} [i_1(\theta) - i_2(\theta)]^2 d\theta$$
$$= \frac{r_{Son}}{2\pi}\int_\pi^{2\pi}\left[i_1(\theta) - \frac{V_{eq}\sin(\theta+\varphi)}{R_{eq}}\right]^2 d\theta \quad (46)$$

The power loss due to the gate drive can be expressed as [30]

$$P_g = 2r_g\pi^2 f_S^2 C_{iss}^2 V_{gs}^2, \quad (47)$$

where r_g is the gate resistance, C_{iss} is the input capacitance, and V_{gs} is the amplitude of the drive voltage.

The power-conversion efficiency is predicted using (43)–(47) as

$$\eta = \frac{P_O}{P_O + P_{L1} + P_{L2} + P_{L3} + P_{Son} + P_g}, \quad (48)$$

where the output power P_O can be obtained from

$$P_O = \frac{R_L I_O^2}{2}. \quad (49)$$

VIII. Experimental Verification

This section shows the experimental verifications of the proposed power oscillator. We provide the two design examples with 0.8 MHz and 6.78 MHz oscillation frequencies. In the 0.8 MHz power oscillator, a comparison between the experimental results and analytical prediction is performed. Also, in the 6.78 MHz power oscillator, comparative experiments with the proposed and conventional class-E power oscillators are conducted.

A. 0.8 MHz Load-Independent Class-E Power Oscillator

1) Circuit Design: The proposed power oscillator was designed with the specifications of input voltage $V_I = 30$ V and load resistance $R_{Ld} = 35\ \Omega$. Based on the specifications, we selected the IRF530 MOSFET from Vishay Siliconix for the switching device. Considering the rating of the gate-source voltage of the MOSFET, the amplitude of the gate-drive voltage was set to $V_{gs} = 20$ V.

In order to ensure the pure sinusoidal currents in i_2 and i_O, the normalized inductances are specified as $\omega_S L_2/R_{Ld} = 0.6$ and $\omega_S L_3/R_{Ld} = 3$, respectively. Fig. 11 shows the bode plot of the LCLC filter obtained using (20) and (21), where R_{Lr} is specified at 10 Ω. It can be seen that the filter can eliminate the harmonic frequencies, such as 1.6 MHz and 2.4 MHz, which is compatible with the assumption (i) in the analysis. Additionally, the phase shift becomes $-\pi/2$, which is constant for different normalized load resistances at 0.8 MHz.

Table I gives the theoretical, simulation, and measured circuit parameters. The theoretical component values were derived from the design equations presented in Section VI-D. The simulation was conducted using OrCAD PSpice, and the results were in good agreement with the theoretical predictions. In the experiment, the component values were measured

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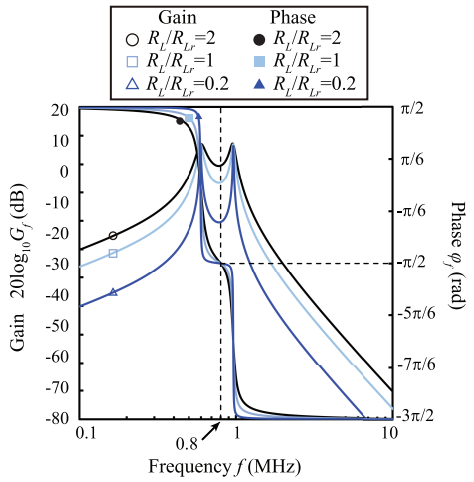


Fig. 11. Voltage gain $20\log_{10}G_f$ and phase shift φ_f of the LCLC filter as functions of the frequency for fixed normalized load resistance.

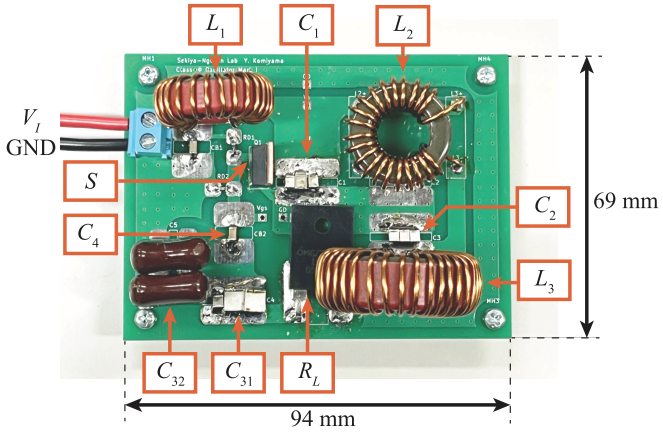


Fig. 12. Photo of the 0.8 MHz load-independent class-E power oscillator.

TABLE I
THE COMPONENT VALUES FOR 0.8 MHZ LOAD-INDEPENDENT CLASS-E POWER OSCILLATOR

Symbol	Theoretical	Simulation	Measured
C_1	9.01 nF	9.01 nF	8.92 nF
C_2	9.47 nF	9.47 nF	9.48 nF
C_{31}	3.50 nF	3.50 nF	3.00 nF
C_{32}	20.53 nF	20.53 nF	19.64 nF
L_1	2.63 μH	2.63 μH	3.70 μH
L_2	4.18 μH	4.18 μH	4.19 μH
L_3	17.41 μH	17.41 μH	17.56 μH
R_1	160 kΩ	160 kΩ	159 kΩ
R_2	20 kΩ	20 kΩ	20 kΩ
f_S	0.8 MHz	0.8 MHz	0.8 MHz
η	94.2 %	93.6 %	93.8 %
P_O	50.3 W	50.6 W	49.5 W

by the E4990A Impedance Analyzer from Keysight Technologies. To implement the inductors, we used Mix 2 Iron-Powder Core from Micrometals. The core sizes are selected considering the peak magnetic flux density. We adopted T94-2 for L_1 and L_2, and T130-2 for L_3. According to the datasheet, the expected peak magnetic flux densities of L_1, L_2, and L_3 are calculated to be 6.9 mG, 4.6 mT, and 6.0 mT, respectively. The permeability variation of Mix 2 material for these peak magnetic flux densities is less than 0.5 %, which is sufficiently small. The capacitor C_{32} was implemented with the Dip Mica Capacitor from Nitsuko Electronics. Fig. 16 shows the prototype power oscillator.

2) Operating Waveform: Fig. 13 shows the superimposed experimental and analytical waveforms of the load-independent class-E power oscillator, where the experimental waveforms were measured with the MSO58B Mixed Signal Oscilloscope from Tektronix. It can be seen from Fig. 13(a)-(c) that the experimental waveforms are in good agreement with the analytical one, demonstrating the validity of the experiment. In the waveforms, the switch voltage achieves the ZVS regardless of the load resistance. Additionally, the amplitudes of the gate-drive voltage and output current remain constant against load variations. Therefore, we can confirm the load-independent operation of the proposed power oscillator from the experiment.

Fig. 14 shows the superimposed experimental waveforms of the load-independent class-E power oscillator against component tolerances, where subscripts 'r' mean their rated component values. To confirm the sensitivity to component tolerances, we varied L_2, C_2, and L_3 for $\pm$10%. It can be seen from Fig. 14(a) that the variations in L_2 did not significantly affect the ZVS and output current. This is because the power oscillators inherently track the changes in the resonant frequency. Indeed, the oscillation frequencies for L_2/L_{2r} = 1.1 and L_2/L_{2r} = 0.9 were 0.789 MHz and 0.804 MHz, respectively. We can also see from Fig. 14(b) and (c) that the proposed power oscillator maintains stable operation even with the component tolerances in C_2 and L_3. It is particularly noteworthy that the power oscillator demonstrates robustness against variations in L_3, despite the load network having a high quality factor of $Q_3 \approx 10$ for R_L/R_{Lr} = 1. The oscillation frequencies for C_2/C_{2r} = 1.1, C_2/C_{2r} = 0.9, L_3/L_{3r} = 1.1, and L_3/L_{3r} = 0.9 were 0.798 MHz, 0.801 MHz, 0.772 MHz, and 0.823 MHz, respectively. From these results, we can state that the proposed power oscillator maintains output power and efficiency by automatically changing the frequency against component tolerances.

3) Circuit Characteristic: Fig. 15(a) shows the output current amplitude and power-conversion efficiency as functions of the normalized load resistance. The power-conversion efficiency, including the gate-drive losses, was calculated as

$$\eta = \frac{P_O}{P_I} = \frac{R_L I_O^2}{2 V_I I_I}, \tag{50}$$

where P_I and I_I are the input power and current, respectively. It can be seen that the output current keeps a constant amplitude against load resistance. Besides, the high power-conversion efficiency is maintained due to the load-independent ZVS. The experiemntal maximum efficiency 93.8 % was obtained at R_L/R_{Lr} = 1.5. The experimental results and analytical prediction agree quantitatively, which supports the validity of the experiment.

Fig. 15(b) shows the power-loss breakdown as a function of the normalized load resistance, which is obtained from the analysis shown in Section VII. It can be seen that the

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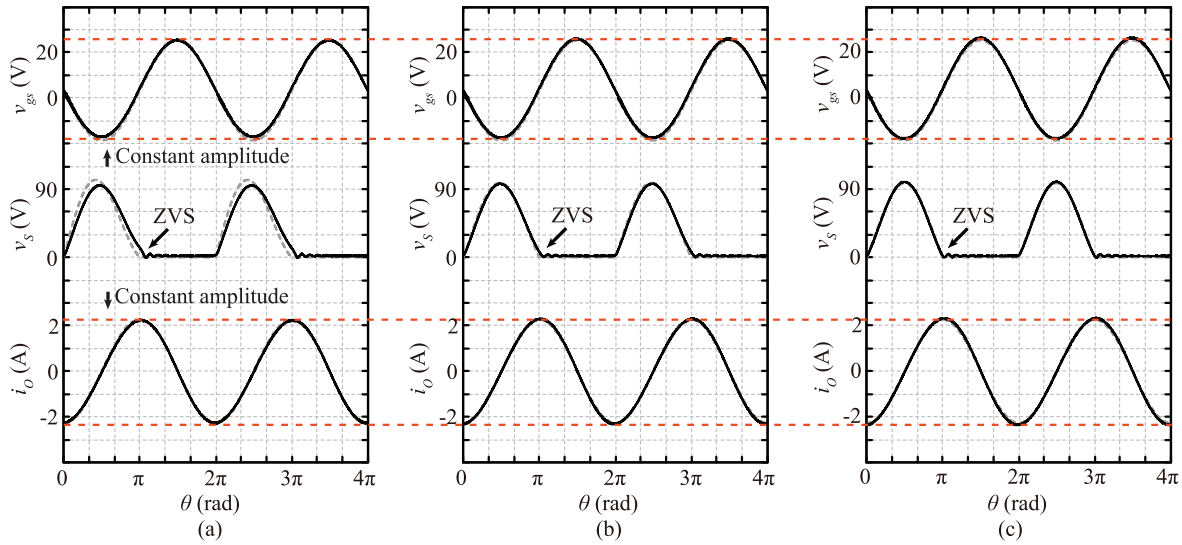


Fig. 13. Superimposed experimental and analytical waveforms of the load-independent class-E power oscillator for fixed load resistance. (solid line: experiment, dashed line: analytical.) (a) $R_L/R_{Lr} = 1$. (b) $R_L/R_{Lr} = 0.5$. (c) $R_L/R_{Lr} = 0$ (short circuit load).

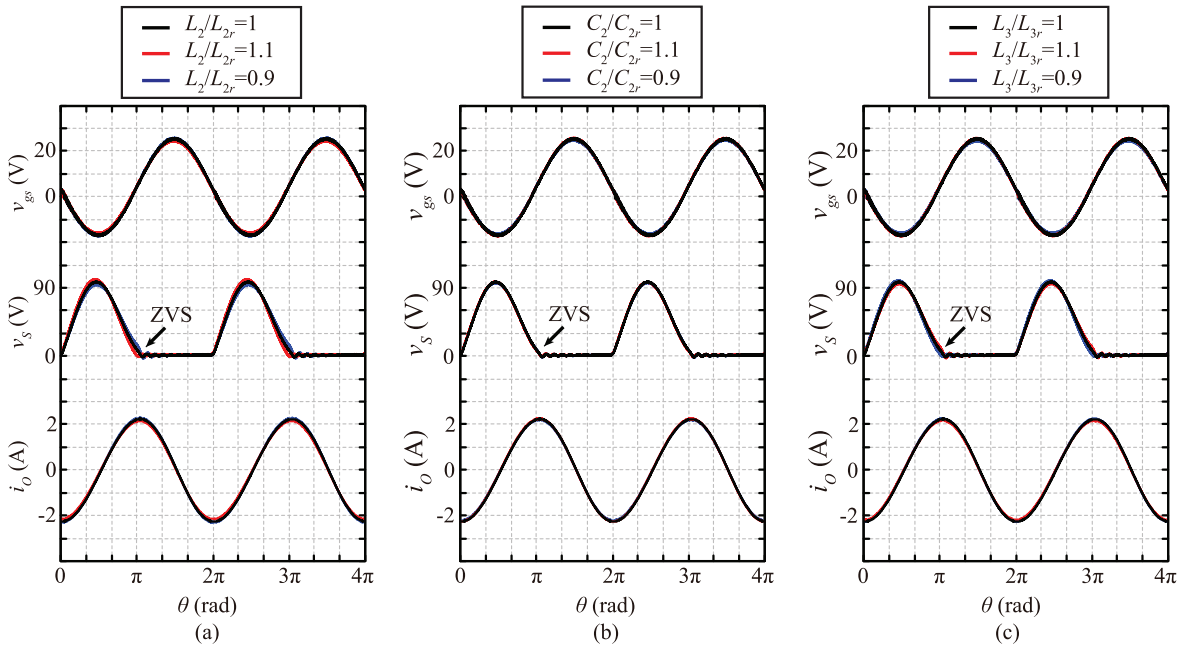


Fig. 14. Superimposed experimental waveforms of the load-independent class-E power oscillator against component tolerances for $R_L/R_{Lr} = 1$. (a) For fixed L_2. (b) For fixed C_2. (c) For fixed L_3.

power loss caused by the switching device is the dominant loss factor. The conduction loss in the switching device becomes especially large in heavy load conditions. The power loss that occurs in the ESR of inductor L_3 is kept constant against load resistance. This is because the amplitude I_O in (45) is constant due to the load-independent characteristics.

B. 6.78 MHz Power Oscillators

1) Circuit Design: The design specifications are given as input voltage $V_I = 25$ V and load resistance $R_{Ld} = 50\ \Omega$. Based on the specifications, we selected GS61004B e-GaN HEMT from Infineon Technologies for the switching device. From the datasheet, the maximum gate-to-source voltage and the threshold voltage were obtained as $V_{Sm} = 7$ V and $V_{th} = 1.7$ V, respectively. Thus, the amplitude of the gate-drive voltage was designed to be $V_{gs} = 4$ V. The normalized inductances are given as $\omega_S L_2/R_{Ld} = 0.7$ and $\omega_S L_3/R_{Ld} = 5$, respectively. From these specifications, the component values of the proposed power oscillator are determined as given in Table II. In addition, we designed the conventional class-E power oscillator in Section II with the same specifications, whose component values are also given in Table II. Fig. 16 shows the prototype power oscillator. The Mix 6 Iron-Powder Cores, which have lower permeability than the Mix 2 material, were employed to implement the inductors. Additionally, all

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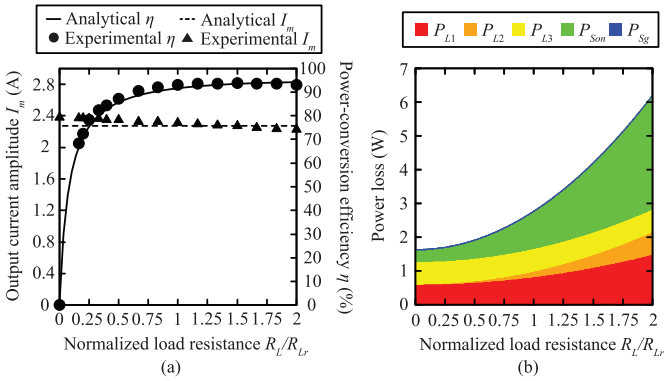


Fig. 15. Circuit characteristics of the proposed power oscillator as functions of the normalized load resistance. (a) Output-current amplitude I_m and power-conversion efficiency η. (b) Power-loss breakdown.

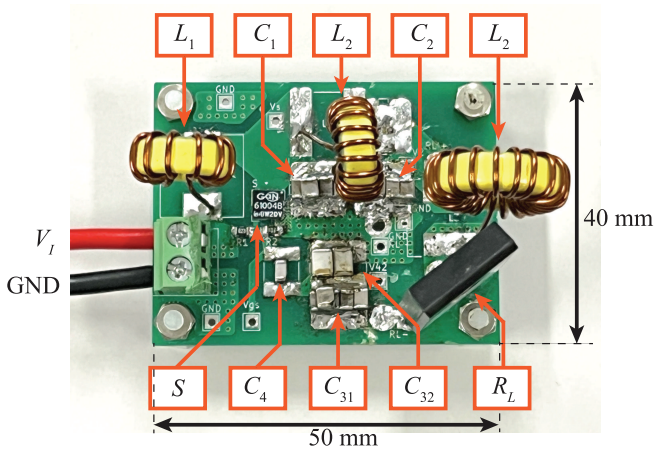


Fig. 16. Photo of the 6.78 MHz load-independent class-E power oscillator.

TABLE II
THE MEASURED COMPONENT VALUES FOR 6.78 MHZ POWER OSCILLATORS

Conventional		Proposed	
Symbol	Value	Symbol	Value
C_1	82 pF	C_1	517 pF
C_2	246 pF	C_2	652 pF
C_3	100 pF	C_{31}	389 pF
C_4	2048 pF	C_{32}	6478 pF
L_1	22 μH	L_1	0.64 μH
L_2	3.44 μH	L_2	1.02 μH
L_3	2.04 μH	L_3	2.35 μH
R_1	120 kΩ	R_1	120 kΩ
R_2	10 kΩ	R_2	10 kΩ

the capacitors were implemented with multilayer ceramic capacitors.

2) Experimental Results for Class-E Power Oscillator: Fig. 17 shows the experimental waveforms of the conventional class-E power oscillator. We can see from Fig. 17 that the gate-drive voltage v_{gs} did not shape a pure sinusoid, causing a voltage depression at the turn-OFF period. A similar observation has also been reported in [9] and is believed to be caused by the nonlinearity of the gate parasitic capacitance. As widely recognized, the allowable gate-voltage range of GaN e-HEMT is extremely narrow. Therefore, this operation could potentially damage the device. We can also observe that the switch voltage achieved the ZVS for $R_L/R_{Lr} = 1$, as shown in Fig. 17(a). Although we tried to experiment with the operation for the case of $R_L/R_{Lr} < 1$ a number of times, the waveforms could not be measured even at $R_L/R_{Lr} = 0.9$ due to the instantaneous damage to the switching device. This is because a sudden reverse conduction occurred in the GaN e-HEMT at the turn-OFF period, which harmed the device. Therefore, we measured the operation for $R_L/R_{Lr} = 2$, as shown in Fig. 17(b). However, the switch voltage could not achieve the ZVS. Besides, the amplitude of the output current changed depending on the load resistance.

The experimental results show that the conventional class-E power oscillator cannot tolerate even small load variations, which significantly degrades the overall performance.

3) Experimental Results for Proposed Power Oscillator: Fig. 18 shows the superimposed experimental and analytical waveforms of the load-independent class-E power oscillator. We can see from Fig. 18 that the gate-drive voltage was purely sinusoid without the voltage depression compared with the conventional class-E power oscillator in Fig. 17. This is because the nonlinear gate capacitance was absorbed into the external capacitance C_{32}, mitigating the effect of the nonlinearity. We can also see that the ZVS was achieved regardless of the load resistance. Additionally, the output current maintained constant amplitude and phase shift. Moreover, the load-independent operation contributes to generating gate-drive voltage with constant amplitude and phase shift against load variations. Therefore, the proposed power oscillator maintained a stable oscillation against load variations. The experimental and analytical waveforms were in good agreement, which demonstrated the validity of the experiment.

4) Circuit Characteristic: Fig. 19 illustrates the output-current amplitude and efficiency as functions of load resistance for conventional and proposed power oscillators, where I_{mr} is the output-current amplitude in the rated condition. It can be seen from Fig. 19 that the proposed power oscillator maintained a constant output-current amplitude against load variations. Additionally, the high power-conversion efficiency above 80 % was maintained within $0.4 \le R_L/R_{Lr} \le 2$ due to the load-independent ZVS operation. The peak power-conversion efficiency was 87.4 % at $R_L/R_{Lr} = 1.6$.

The conventional power oscillator could not oscillate within $R_L/R_{Lr} < 1$ due to the instantaneous damage of the switching device. Therefore, the circuit performance for $1 \le R_L/R_{Lr} \le 2$ was evaluated. The output current fluctuated compared with the proposed power oscillator. Besides, the power-conversion efficiency was up to 82.8 % and decreased with the increase of load resistance. The reasons for the efficiency deterioration are considered to be the switching loss due to non-ZVS and growing gate-drive loss in (47) by the increase in the amplitude of the gate-drive voltage.

From these results, we validated the effectiveness of the load-independent operation of the proposed power oscillator.

C. Discussion for Scalability

The design equations provided in this paper work for any specifications in power and frequency. Using the design

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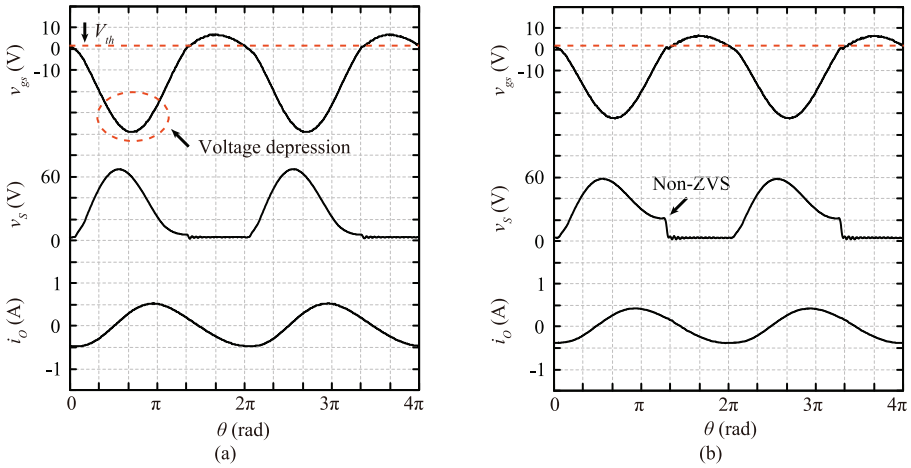


Fig. 17. Experimental waveforms of the conventional class-E power oscillator for fixed load resistance. (a) $R_L/R_{Lr} = 1$. (a) $R_L/R_{Lr} = 2$.

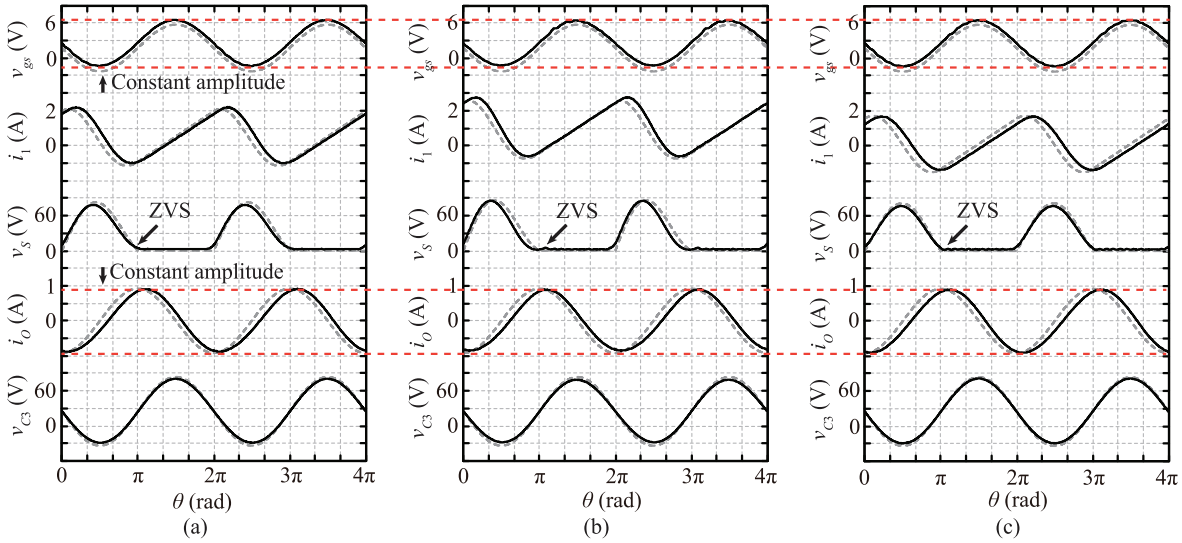


Fig. 18. Superimposed experimental and analytical waveforms of the load-independent class-E power oscillator for fixed load resistance. (solid line: experiment, dashed line: analytical.) (a) $R_L/R_{Lr} = 1$. (a) $R_L/R_{Lr} = 2$. (c) $R_L/R_{Lr} = 0.2$.

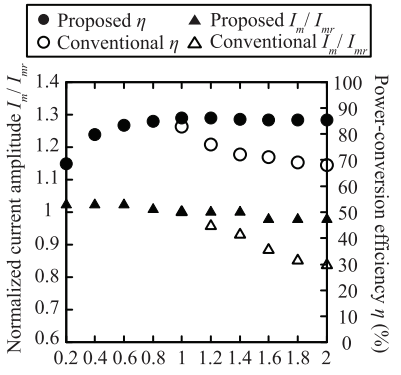


Fig. 19. Normalized output-current amplitude I_m/I_{mr} and power-conversion efficiency η as functions of normalized load resistance R_L/R_{Lr}.

equations, the proposed power oscillator can be applied to various scenarios. Meanwhile, the proposed power oscillator has potential challenges in scalability when considering practical implementation.

1) Scalability for Power: One challenge in scaling up the proposed power oscillator for higher power lies in the implementation of capacitor C_{32}. The design value of C_{32} is given by (41). This equation indicates that C_{32} is proportional to V_I/V_{gs}. The amplitude of the gate-drive voltage V_{gs} is typically low. In contrast, input voltage V_I becomes high in high-power applications. Accordingly, the ratio V_I/V_{gs} becomes large, necessitating a large capacitance for C_{32}. However, large capacitors that are stable above MHz frequencies are limited.

To address this issue, we introduce a modified circuit configuration. Fig. 20 shows a modified schematic of the proposed power oscillator. In this circuit configuration, the resistors R_3 and R_4 are introduced instead of the capacitor C_{32}. The amplitude of the gate-drive voltage can be adjusted using sufficiently large resistors R_3 and R_4 without large capacitors.

2) Scalability for Frequency: The gate-drive method is one of the challenges in achieving high-frequency and high-efficiency operation. The proposed power oscillator has an advantage in absorbing the gate capacitance. However, the

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TABLE III
COMPARISON WITH THE POWER OSCILLATORS

	Year	Topology	Switching condition	Robustness against load variations	Operating frequency	Output power	Power-conversion efficiency
[5]	2021	Class-E	ZVS	No	0.786 MHz	1.1 W	89.5 %
[6]	2023	Class-E	ZVS	No	0.792 MHz	0.80 W	71.6 %
[7]	2023	Push-pull class-E	ZVS	No	3.3 MHz	30 W	89.3 %
[9]	2019	Class-E	ZVS	No	0.8 MHz	0.96 W	89.0 %
[10]	2020	Class-E	ZVS	No	0.8 MHz	1.0 W	92.0 %
[11]	2022	Class-E	ZVS	No	6.78 MHz	N/A	N/A
[12]	2018	Class-E/F_3	ZVS	No	0.8 MHz	0.91 W	86.2 %
[13]	2024	Class-E/F_3	ZVS	No	0.8 MHz	1.0 W	93.5 %
[14]	2024	Class-Φ_3	ZVS	No	1 MHz	70 W	92.2 %
[16]	2023	Class-E	ZVS	Yes	13.56 MHz	0.34 W	85.0 %
[18]	2020	Class-E^{-1}	ZCS	Yes	1 MHz	N/A	N/A
[19]	2021	Class-E^{-1}	ZCS	Yes	1 MHz	N/A	N/A
[20]	2022	Class-E^{-1}	ZCS	Yes	1 MHz	20 W	76.5 % (DC/DC)
[21]	2023	Class-E^{-1}	ZCS	Yes	1 MHz	12 W	95.0 %
This work	-	Class-E	ZVS	Yes	0.8 MHz / 6.78 MHz	50 W / 26 W	93.8 % / 87.4 %

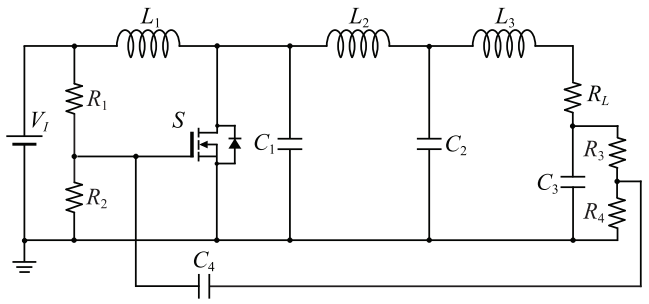


Fig. 20. Modified circuit schematic of the proposed power oscillator using divided resistors.

sinusoidal gate voltage has long rising and falling times, which can increase switching losses.

The driver IC with a square waveform can reduce the switching transition time. By employing the driver IC, the power losses in the main switch are reduced. However, square-wave driving causes significant hard-switching losses in the driver IC [31]. Furthermore, propagation delays caused by the driver IC disturb the phase-shift requirement of the power oscillator. Therefore, an additional consideration for adjusting the phase shift is required.

In high-frequency operation, an appropriate gate-drive design that considers these trade-offs should be selected.

IX. COMPARISON WITH CONVENTIONAL WORKS

Table III gives the comparison with the state-of-the-art power oscillators. In [5] and [6], choke-less class-E power oscillators have been proposed. These power oscillators eliminate the need for choke inductors and DC-blocking capacitors. Consequently, the power density of the circuit can be increased. In [7], a push-pull class-E power oscillator has been proposed. The push-pull topology enhances the power density by increasing the power-output capability. For a similar purpose, the class-E/F_3 and the class-Φ_3 power oscillators have been proposed [12], [14]. These power oscillators increase power-output capability using a harmonic resonant filter. However, all these works focus on the circuit operation only in the rated condition. In real-world scenarios of power electronics, component tolerances, and load variations are essential factors to consider.

To address the component tolerances, the self-tuned class-E and class-E/F_3 power oscillators [9], [10], [13] have been proposed. These works have clarified that the power oscillators can track the deviations in the resonant frequency by employing a low-Q feedback network. In [11], a transmission line stub has been proposed to enhance the stability of the oscillation against component deviations. Although the researches in [9], [10], [11], and [13] consider the deviations of inductors and capacitors, they cannot achieve soft switching against a wide range of load variations.

To overcome the sensitivity against load variations, an injection-locked class-E power oscillator has been proposed [16]. In [16], the injection-locked power oscillator is used for the transmission circuit of the Wireless Power Transfer (WPT) system. The injection locking allows high-efficiency operation at a fixed frequency against the coupling factor deviations. In [18], [19], [20], and [21], the load-independent class-E^{-1} power oscillators have been proposed. The load-independent class-E^{-1} power oscillators ensure the ZCS regardless of the load resistance. Meanwhile, the ZCS power oscillators cannot achieve ZVS. The non-ZVS at multi-MHz frequency causes serious efficiency deterioration, high switch-voltage stress, and unstable oscillation. In summary, no power oscillator has been proposed that can consistently maintain ZVS under load variations.

The proposed power oscillator is the first load-independent class-E ZVS power oscillator. The proposed power oscillator ensures ZVS and constant amplitudes in output current and gate-drive voltage against load variations. Additionally, the proposed circuit topology inherently absorbs the gate-source capacitance, making it suitable for high-frequency operation.

X. CONCLUSION

This paper has introduced the load-independent class-E ZVS power oscillator. The proposed power oscillator inherently achieves ZVS irrespective of the load resistance. Furthermore, the amplitude and phase shift of both the gate-drive voltage and the output current remain unaffected by the load

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resistance. These characteristics allow the circuit to operate with high stability and efficiency across a wide range of load variations. From the circuit analysis, the load-independent conditions, optimal design point, and design equations are obtained. Furthermore, the power-loss analysis provides the loss breakdown and efficiency prediction. The experimental verifications were carried out under 0.8 MHz and 6.78 MHz oscillation frequencies. The prototype power oscillator demonstrates high power-conversion efficiency and stability against load variations. The quantitative agreement between the experiment and analysis showed the validity of the experimental results.

The application of WPT systems is one of the promising research directions for power oscillators. Future studies should address the potential challenges of power oscillators for considering real-world scenarios. For example, the enhancement of stability, the operation within a narrow Industrial, Scientific, and Medical (ISM) frequency band, and the scalability for power and frequency.

APPENDIX

We provide the detailed analytical procedure for deriving the waveform equations of (1) and (2).

During the turn-OFF state, the differential equation for the switch voltage v_S is obtained from Kirchhoff's Voltage Law (KVL) as

$$\frac{d^2 v_S(\theta)}{d\theta^2} + \omega_1^2 v_S(\theta) = \omega_1^2 \left[V_I - \frac{\omega_S L_1 V_{eq} \cos(\theta + \varphi)}{R_{eq}} \right]. \quad (51)$$

The solution to (51) yields the expression for the switch voltage in the turn-OFF period as

$$v_S(\theta) = \alpha \cos(\omega_1 \theta) + \beta \sin(\omega_1 \theta) + V_I - \frac{\omega_S L_1 \omega_1^2 V_{eq} \cos(\theta + \varphi)}{R_{eq}(\omega_1^2 - 1)}. \quad (52)$$

From KVL, we obtain

$$\frac{di_1(\theta)}{d\theta} = \frac{1}{\omega_S L_1} [V_I - v_S(\theta)]. \quad (53)$$

By substituting (2) into (53) and solving the differential equation, we obtain the expression of current i_1 during turn-OFF period as

$$i_1(\theta) = -\frac{\omega_1 [\alpha \sin(\omega_1 \theta) - \beta \cos(\omega_1 \theta)]}{\omega_S L_1} + \frac{\omega_1^2 V_{eq} \sin\theta}{R_{eq}(\omega_1^2 - 1)} \quad (54)$$

During the turn-ON state, the switch is shorted out, making the switch voltage to be zero as

$$v_S(\theta) = 0. \quad (55)$$

From KVL, we obtain

$$\frac{di_1(\theta)}{d\theta} = \frac{V_I}{\omega_S L_1}. \quad (56)$$

By solving (56), we obtain

$$i_1(\theta) = \frac{V_I}{\omega_S L_1}\theta + \gamma, \quad (57)$$

where γ is a constant. By considering the continuity of the current i_1, the boundary condition can be formulated as

$$i_1(\pi^-) = i_1(\pi^+). \quad (58)$$

From (54), (57), and (58), the unknown constant γ can be obtained, and the expression of i_1 during turn-ON period becomes

$$i_1(\theta) = \frac{V_I(\theta - \pi) - \omega_1[\alpha \sin(\pi\omega_1) - \beta \cos(\pi\omega_1)]}{\omega_S L_1}, \quad (59)$$

Also, the expressions of α and β in (5) and (6) are obtained by solving the linear equations of

$$i_1(0) = i_1(2\pi), \quad (60)$$

and

$$\left.\frac{di_1(\theta)}{d\theta}\right|_{\theta=0} = \left.\frac{di_1(\theta)}{d\theta}\right|_{\theta=2\pi}. \quad (61)$$

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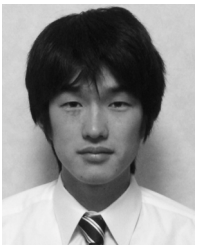
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